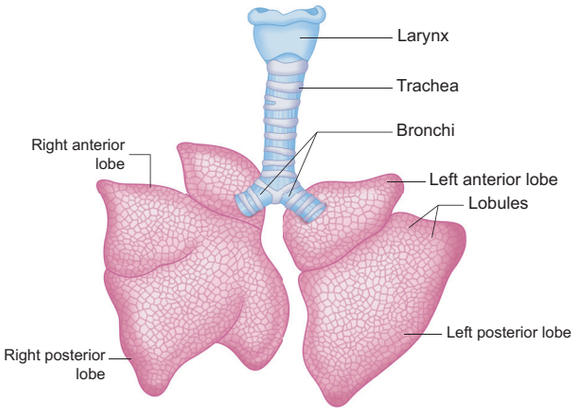


Fig. 13.8 Lungs of Rabbit

Each lung is enclosed by a double membranous **pleura**. Atmospheric air passes through the external nostril and nasal passages into the pharynx. From the pharynx it passes through the glottis into the wind pipe.

The anterior part of the wind pipe is enlarged to form the **larynx** or **voice box** with its wall supported by four cartilaginous plates. Inside the larynx lies the vocal cord and its vibrations result in the production of sound. The larynx leads into **trachea** or **wind pipe**.

Tracheal walls are supported by rings of cartilage which help in the free passage of air. The **epiglottis** prevents the entry of food into the trachea through the glottis. The trachea divides into two branches called the **bronchi** one entering into each lung and dividing into further branches called **bronchioles** which end in alveoli.

The respiratory events consist of **inspiration** (breathing in) and **expiration** (breathing out) allowing exchange of gases (oxygen and carbon dioxide). Inspiration is an active process while expiration is a passive process.

13.2.6 Circulatory System

The circulatory system is formed of blood, blood vessels and heart. The heart is pear shaped and lies in the thoracic cavity in between the lungs. It is enclosed by **pericardium**, a double layered membrane.

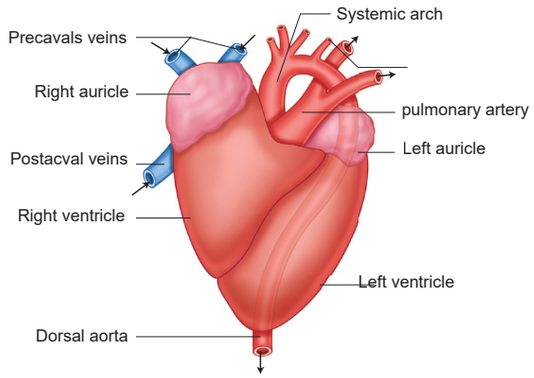


Fig. 13.9 Heart of Rabbit - Ventral View

The heart is four chambered with two auricles and two ventricles. The right and left auricles are separated by **interauricular septum**, similarly right and left ventricles are separated by **interventricular septum**.

The right auricle opens into the right ventricle by right **auriculoventricular aperture**, guarded by a **tricuspid valve**. The left auricle opens into the left ventricle by left auriculoventricular aperture guarded by a **bicuspid valve** or **mitral valve**. The opening of the pulmonary artery and aorta are guarded by three **semilunar valves**.

The right auricle receives deoxygenated blood through two **precaval** (superior vena cava) and one **postcaval** (inferior vena cava) veins from all parts of the body. The left auricle receives oxygenated blood from the pulmonary veins from the lungs. From the right ventricle arises pulmonary trunk which carries the deoxygenated blood to the lungs and from the left ventricle arises the systemic arch (aorta) which supplies oxygenated blood to all parts of the body.

13.2.7 Nervous System

The nervous system in rabbit is formed of the central nervous system (CNS), peripheral nervous system (PNS) and autonomic nervous system (ANS).

CNS consists of brain and spinal cord. PNS is formed of 12 pairs of cranial nerves and 37 pairs of spinal nerves. ANS comprises sympathetic and parasympathetic nerves.

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